

Linea Tokens Audit Summary

Prepared by Cyfrin

Version 2.5

Lead Auditors

Immeas

Chinmay

September 10, 2025

Contents

1 About Cyfrin **2**

2 About This Document **2**

3 Disclaimer **2**

4 Copyright and License **2**

5 Audit Overview **3**

6 Audit Scope **3**

7 Deployment Verification **3**

8 Risk Classification **4**

9 Protocol Summary **4**

9.1 Key Components 4

10 Attestation **5**

1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Linea on Linea Tokens. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.5 dated September 10, 2025, was delivered to Linea, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

© 2026 Cyfrin Inc. All rights reserved. Material from Linea or third parties remains the property of its owners.

This summary may be shared only in complete and unmodified form, so that any copy can be checked against the hash in its attestation. Short quotations with attribution to Cyfrin are permitted, provided they do not misrepresent this summary's scope or results. Except as permitted by applicable law, no part of this summary may otherwise be copied, modified, adapted, sold or sublicensed, or used to create derivative works, without Cyfrin's prior written permission.

Cyfrin and the Cyfrin logo are trademarks of Cyfrin Inc. and may not be used to suggest that Cyfrin has reviewed or endorses anything outside this summary's scope, or to misrepresent its results.

5 Audit Overview

Over the course of 3 days, the Cyfrin team conducted an audit on the Linea Tokens code provided by Linea. The findings consist of 5 Informational findings.

Summary

Project Name	Linea Tokens
Repository	linea-tokens
Commit	44640f0965a5...
Fix Commit	91036daf2331...
Audit Timeline	Jul 25th - Jul 29th, 2025
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	0	0	0
Medium Risk	0	0	0
Low Risk	0	0	0
Informational	5	2	3
Gas Optimizations	0	0	0
Total	5	2	3

6 Audit Scope

```
src/airdrops/TokenAirdrop.sol
src/L1/LineaToken.sol
src/L2/L2LineaToken.sol
```

7 Deployment Verification

All updates, including those addressing the Cyfrin audit findings, are contained in PR#20. Cyfrin validated the final merge commit 91036da.

The bytecode in the `exported-artifacts` folder was confirmed to match a fresh compile using the settings in `foundry.toml`. The deployed bytecode of the `LineaToken` implementation on Ethereum and the `L2LineaToken` implementation on Linea was confirmed to match the bytecode committed in `exported-artifacts`:

- `LineaToken` implementation (Ethereum): 0xe03f157de67ac4b2a9a949d64d2a3c64ffa1bc55
- `L2LineaToken` implementation (Linea): 0xe03f157de67ac4b2a9a949d64d2a3c64ffa1bc55

Both implementations are pointed to by the proxy at `0x1789e0043623282d5dcc7f213d703c6d8bafbb04` (Ethereum, Linea).

Later the code was moved to the public Linea monorepo in commit a9a43aa. Cyfrin again verified that the code, the compiled contracts (using `foundry.toml`) and `exported-artifacts` all match what is compiled and committed in 91036da.

SHA-256 digests at monorepo@a9a43aa:

```
contracts-tge/exported-artifacts/L2LineaToken.json
  4bdfbbb35d50f9ef8cb982d4100628f4ff60fc07b6c4c32daae8e80cc1273664
contracts-tge/exported-artifacts/LineaToken.json
  17a7f64540f4fa406889fc9bf3ea73d2e21cb7b272d2152073420d344c9db0f9
contracts-tge/exported-artifacts/TokenAirdrop.json
  1ab698b5f166d6b49d0b21567d0f75193bd50db5118f5b9a955bebf5b9f86321
```

SHA-256 digests at linea-tokens@91036da:

```
exported-artifacts/L2LineaToken.json
  4bdfbbb35d50f9ef8cb982d4100628f4ff60fc07b6c4c32daae8e80cc1273664
exported-artifacts/LineaToken.json
  17a7f64540f4fa406889fc9bf3ea73d2e21cb7b272d2152073420d344c9db0f9
exported-artifacts/TokenAirdrop.json
  1ab698b5f166d6b49d0b21567d0f75193bd50db5118f5b9a955bebf5b9f86321
```

8 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

9 Protocol Summary

The Linea Token is an ERC-20 deployed by the Linea team on L1 (Ethereum) and bridged to L2 (Linea chain) for distribution via `TokenAirdrop`. Bridged tokens are seeded into the airdrop contract, which computes allocations based on three soulbound factor tokens, `PRIMARY_FACTOR_ADDRESS`, `PRIMARY_CONDITIONAL_MULTIPLIER_ADDRESS`, and `SECONDARY_FACTOR_ADDRESS`. Going forward, Linea Token will also serve as the on-chain governance token across the Linea ecosystem.

9.1 Key Components

- **TokenAirdrop:**
  - Uses three soulbound tokens to calculate each account's allocation:
    1. **Primary factor** (`PRIMARY_FACTOR_ADDRESS`): base balance (18 decimals)
    2. **Conditional multiplier** (`PRIMARY_CONDITIONAL_MULTIPLIER_ADDRESS`): multiplier balance (9 decimals), applied to the primary balance via `× multiplier ÷ DENOMINATOR`
    3. **Secondary factor** (`SECONDARY_FACTOR_ADDRESS`): additional balance (18 decimals)
  - Users claim their allocation via `claim()`.
- **LineaToken (L1):**
  - Upgradeable ERC-20 with 18 decimals, burnable, permit-enabled.
  - The initial supply is bridged to L2.

  - `syncTotalSupplyToL2()` sends `(block.timestamp, totalSupply)` to L2 via the configured `IMessageService`.

- **L2LineaToken (L2):**
  - Upgradeable ERC-20 with 18 decimals, permit (`ERC20PermitUpgradeable`), voting (`ERC20VotesUpgradeable`), and cross-chain sync (`MessageServiceBase`).
  - Bridged between L1 and L2 through the canonical token bridge.
  - `syncTotalSupplyFromL1` receives the L1 total supply relayed by the message service.

10 Attestation

Cyfrin attests that it performed a security review of the Linea Tokens code provided by Linea.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2025-09-10-cyfrin-linea-tokens-v2.5.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Linea has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=6ab2fabbc6ab50e789a7b9663bdad8e054ae946ba9b2fd6de58bafe87307ce7b
F=2025-09-10-cyfrin-linea-tokens-v2.5-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.